MODULE 24 · DOMAIN 5 - DATA COLLABORATION

Secure sharing, listings, clean rooms and native apps

Four ways to collaborate, and none of them copy the data.

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The data never moves. The consumer reads your micro-partitions, pays for their own compute, and can never write.

What can and cannot be shared

SHAREABLE	NOT SHAREABLE
Tables	Non-secure views
External tables and dynamic tables	Stored procedures
Secure views	File stages
Secure materialized views	Tasks
Secure UDFs	Streams
Iceberg tables	

ON THE EXAM

Secure is the theme on the left. A non-secure view leaks its definition and can leak the rows it filters out, which is why it is barred.

Building a share, and what the consumer does

- **CREATE SHARE**, then grant to it: **USAGE on the database**, **USAGE on each schema**, and **SELECT on the objects**.
- Requires **ACCOUNTADMIN** or the **CREATE SHARE** global privilege.
- The consumer creates a **database from the share**. Their administrator grants **IMPORTED PRIVILEGES**, which opens everything in it.
- For finer control, the provider puts **database roles** in the share. The consumer then grants each one to their own account roles.
- Object identifiers must be **uppercase and alphanumeric**, so consumers need no double-quoted identifiers.

> ON THE EXAM
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> The grant pattern is module 8's containment rule again: USAGE on the database, USAGE on the schema, then the object privilege.

Direct share or listing?

DIRECT SHARE

Same region, same cloud

- Shared straight to named consumer accounts
- Only within one region and cloud platform
- No metadata, no discovery, no usage metrics
- Immediate. Nothing is replicated

LISTING

Any region, any cloud

- Cross-Cloud Auto-Fulfillment replicates for you
- Private to named accounts, or public on Marketplace
- Carries metadata, sample queries, usage metrics
- The provider pays storage and replication

THE TIEBREAKER

This is the distinction that decides most sharing questions. A direct share cannot cross a region boundary, full stop.

Reader accounts, exchanges and the Marketplace

- A **reader account** shares with a consumer who has **no Snowflake account**. The **provider creates and pays for it**.
- **Snowflake Marketplace** is the public catalogue. Listings can be free, trial, or paid.
- A **Data Exchange** is a private, managed group where you control who may publish and who may consume.
- To share across regions *without* a listing you need **replication**, which is **blocked if the database contains external tables**.
- **VPS accounts cannot use the Marketplace or public listings**, from module 2. VPS is not a strict superset.

> ON THE EXAM
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> Reader accounts are the answer to 'they are not a Snowflake customer'. The provider pays, which is the detail people forget.

Data Clean Rooms

- Two parties analyse **combined** data without either seeing the other's rows. The classic use is audience overlap.
- The provider links tables and defines **templates**, which are the SQL queries that may be run.
- **Join policies** and **column policies** control which columns may be joined on and which may appear in results.
- **Data in a clean room cannot be queried or viewed directly by either party.** It is reachable only through a template.
- Clean rooms are **versioned native applications**, built on the Native Application Framework.

> ON THE EXAM
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> The signal is 'neither party may see the other's records'. A share cannot do that, because a share grants read access.

The Native Application Framework

- Package logic built from Snowflake primitives, stored procedures, UDFs and shares, and publish it as an **application**.
- The consumer installs it, and **it runs inside their own account**. The logic travels to the data, not the data to the logic.
- Distributed **privately** or on the **Marketplace**, and **versioned**, so consumers can be upgraded.
- The provider's code can be hidden while still operating on the consumer's data.

> ON THE EXAM
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> That inversion is the point. A share moves read access outward; a native app moves executable logic inward.

How this module gets tested

THE REQUIREMENT	THE ANSWER
Share with an account in the same region	Direct share
Share with consumers in other regions or clouds	A listing, with Cross-Cloud Auto-Fulfillment
Share with someone who has no Snowflake account	A reader account, created and paid for by the provider
Share a filtered slice of a table	A secure view. Non-secure views cannot be shared
Neither party may see the other's records	A data clean room
Run our logic on the customer's data, in their account	A native application

ON THE EXAM

Six shapes. The first two are the pair that decides most questions, and the difference is a region boundary.

> A provider in AWS eu-west-1 has a working direct share used by three consumer accounts in the same region. A fourth consumer, in Azure East US, needs the same data.
>
> An engineer adds the fourth account to the existing share. The statement is accepted but the consumer cannot see anything.

What is required?

A The consumer must grant IMPORTED PRIVILEGES before the share becomes visible.

B A direct share works only within one region and cloud. Use a listing with Cross-Cloud Auto-Fulfillment, or set up replication.

C Cross-cloud sharing requires the consumer to upgrade to Business Critical Edition.

D The share must be recreated with the GLOBAL keyword to span cloud platforms.

◆ PAUSE AND DECIDE

What is required?

A Grant IMPORTED PRIVILEGES.

Genuinely required for a consumer to use a share, and it cannot make a share visible across a region boundary.

B Direct shares are same-region. Use a listing or replication.

Documented. Listings with auto-fulfillment replicate the data product to the consumer's region automatically.

C The consumer needs Business Critical.

Sharing is Standard Edition. Business Critical adds failover and private connectivity, not cross-region sharing.

D Recreate the share with GLOBAL.

Invented keyword. There is no syntax that makes a direct share cross regions.

THE SIGNAL

The statement being accepted is the realistic part. Adding the account succeeds; the data simply never reaches another region.

> A provider wants to share a curated dataset with a partner. Their design includes a view that joins three internal tables, a stream on the main fact table so the partner can process changes incrementally, and a task that refreshes a summary.
>
> Only some of these objects can be added to the share.

Which parts fail, and what should be done instead?

A. Only the task fails. Views and streams are both shareable if the provider owns them.

B. All three fail. Only base tables can ever be added to a share.

C. The view fails only if it is not secure, and the stream and task are shareable objects.

D. The view must be secure; the stream and task cannot be shared. Let the partner create their own stream on the shared objects.

◆ PAUSE AND DECIDE

Which parts fail, and what should be done instead?

A Only the task fails.

Streams cannot be shared either, and a non-secure view cannot be shared at all.

B All three fail. Only base tables.

Too restrictive. Secure views, secure materialized views, secure UDFs, external and dynamic tables are all shareable.

C Only the view is a problem.

Half right about the view, and streams and tasks are both on the not-shareable list.

D Secure the view; the partner makes their own stream.

The documented answer, including the specific guidance not to work around it with a secure view on a stream.

THE SIGNAL

The workaround matters. Snowflake explicitly warns against sharing a secure view built on a stream; the consumer makes their own.

> A retailer and a media company want to know how many customers they have in common, and how those customers responded to a campaign. Neither is willing to send the other any customer records, and neither may see the other's identifiers.
>
> An architect proposes that each creates a secure view over hashed email addresses and shares it with the other.

Why is that insufficient, and what fits?

A. A data clean room. Data linked into it cannot be viewed directly by either party, only queried through approved templates.

B. Nothing is wrong. Hashing the identifiers means neither party sees personal data.

C. A reader account for each party, which isolates the data from both organisations.

D. A native application, which is the only way to run analysis across two accounts.

◆ PAUSE AND DECIDE

Why is that insufficient, and what fits?

A A data clean room.

Purpose-built for this. Neither party can query the linked data directly; join and column policies constrain what may be matched and returned.

B Hashing is enough.

A share still grants read access to every row. Hashes can be matched against a known list, so identifiers remain exposed.

C A reader account each.

Reader accounts serve consumers with no Snowflake account. Both parties here are Snowflake customers, and it isolates nothing.

D A native application.

Clean rooms are built on that framework, and the answer named in a scenario like this is the clean room itself.

THE SIGNAL

'Neither may see the other's identifiers' rules out any share, because a share grants read access to rows by definition.

> A consumer has been given a share and can query the tables in it. They now want to enrich the shared customer table with their own internal segment codes and keep the result for reporting.
>
> An engineer attempts an UPDATE against the shared table to add the codes, and it fails.

What should they do?

A. Ask the provider to grant INSERT and UPDATE on the shared objects.

B. Clone the shared database, which produces a writable copy at no storage cost.

C. Create a table in their own account from a query joining the shared table to their internal data.

D. Nothing. Shared data cannot be combined with local data in the same query.

◆ PAUSE AND DECIDE

What should they do?

A. Ask for INSERT and UPDATE.

Shared access is read-only by design. There is no grant that makes a consumer able to write to a provider's table.

B. Clone the shared database.

Cloning requires ownership-level access to the source. A consumer cannot clone objects in a share.

C. Build their own table from a join.

Shared objects can be joined to local data freely. The result is written to the consumer's own table, which they own and pay to store.

D. Shared data cannot be joined locally.

Joining shared and local data is one of the main reasons to consume a share at all.

THE SIGNAL

Read-only is not a limitation to route around. Query out, write locally, and the consumer pays only for what they store.

What to carry into the next module

- **Nothing is copied.** The consumer reads your partitions, pays their own compute, and can only read.
- **Direct share is same region and cloud. A listing crosses both**, via Cross-Cloud Auto-Fulfillment.
- **Non-secure views, stored procedures, stages, tasks and streams cannot be shared.** Let consumers make their own streams.
- **Reader accounts** serve consumers with no Snowflake account, and the **provider pays**.
- **Clean room** when neither party may see the other's rows. **Native app** when your logic must run in their account.

> ON THE EXAM
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> Next: replication and failover, then five scenarios closing Domain 5.